RUBIDIUM STANDARD

A Frequency Standard for an IC 910H

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WHY ?

I already have the Hi Stab Xtal option fitted!

- It seemed a simple and worth while project that would provide stability and frequency readout better than 2 x 10^11.
- It did not need an outside antennae.
- Relatively inexpensive.
- Well researched by Tim Tuck VK2XTT
- The IC 910H has a single reference frequency of 30.200 MHz.

RUBIDIUM STANDARD .. FE 5680A

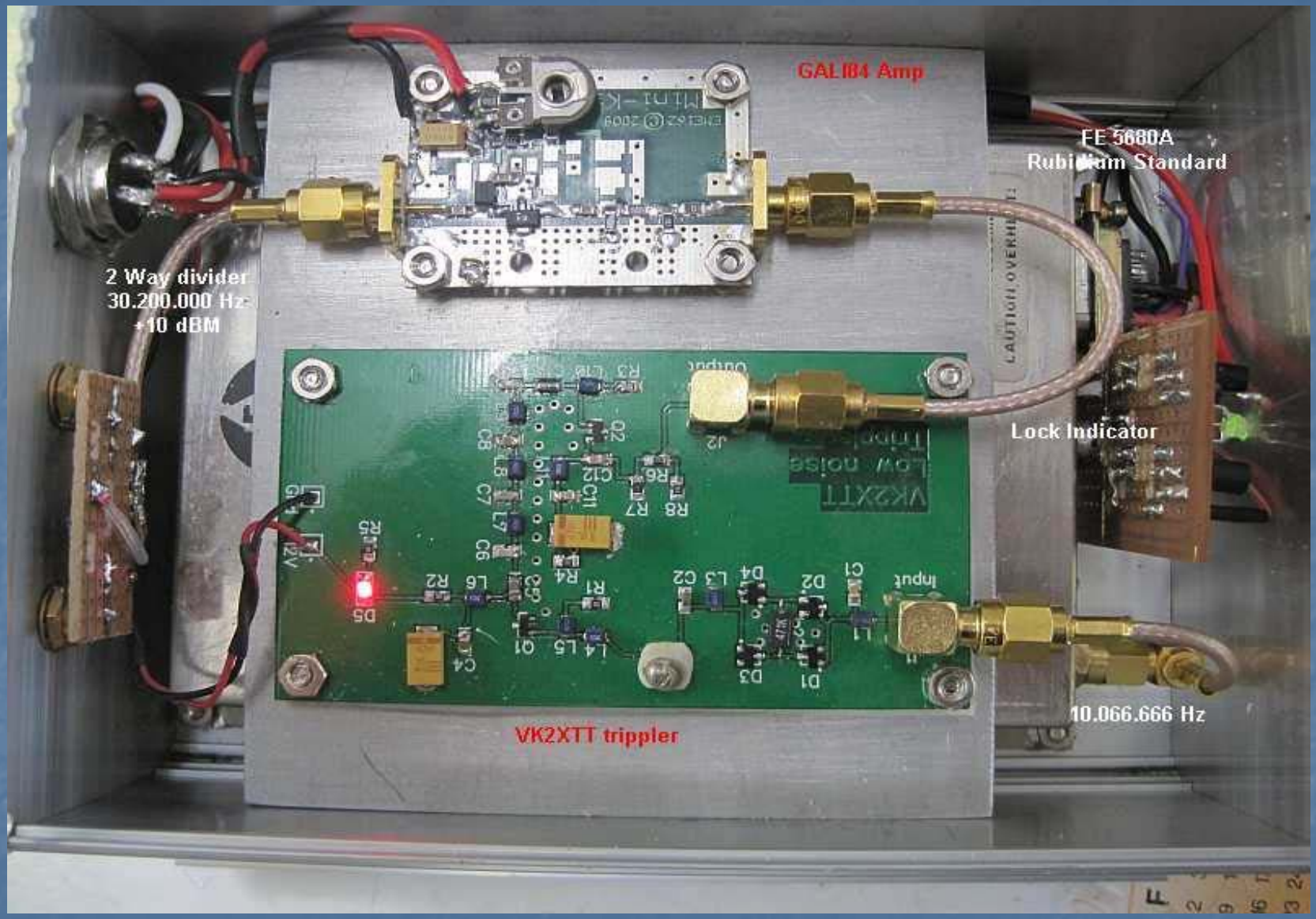
- They are not all the same. You need to obtain one that is fully programmable.
- One that can be programmed to 10.06666 MHz not just the fine offset at 50.255x MHz
- Draw ~ 1.9A on start up falling to about 500mA.
- Run at a case temperature of 47c.
- Use a tripler from 10.0666MHz to 30.200 MHz.
- Additional amplifier to provide 2 x +10dBm 30.2000 MHz outputs. [GALI-39 WIDEBAND AMPLIFIER KIT from Minikits]

Typical programmable frequency response of the FE- 5680A

5 MHz -2.76 dBm second harmonic -39 dBm, 3rd -54 dBm
6 +1.00 dBm second harmonic -38 dBm
7 +4.22 dBm second harmonic -41 dBm
8 +6.59 dBm
9 +7.40 dBm
10 +6.72 dBm
11 +5.33 dBm
12 +3.73 dBm
13 +1.11 dBm spurious at 11.6 @-55 dBm
14 -3.32 dBm
15 -8.73 dBm
16 -14.11 dBm
17 -19.04 dBm
18 -24.09 dBm
19 -28.45 dBm
20 -32.58 dBm .. many outputs 1MHz , 10.3MHz

How it works

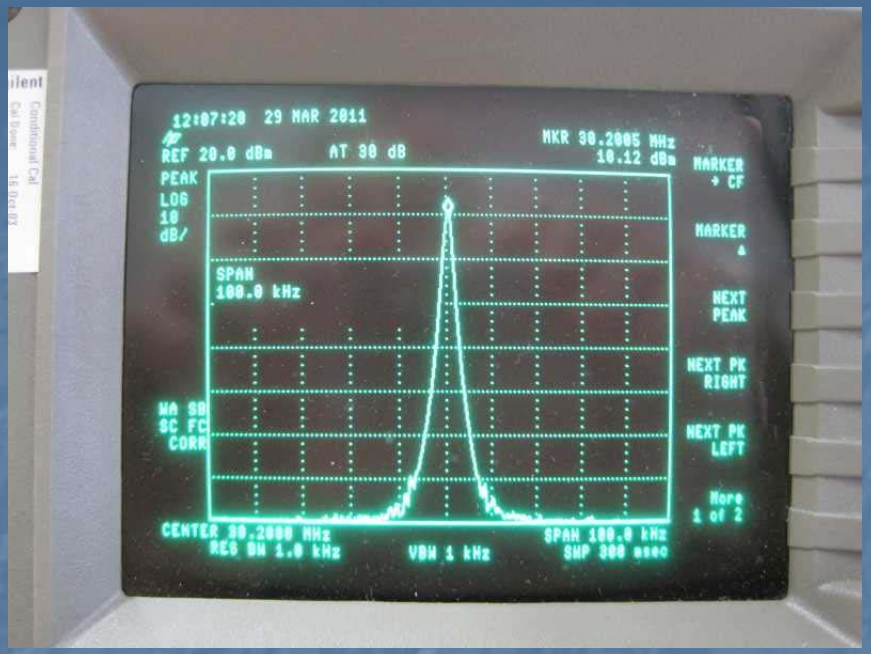
- The Rubidium Physics Package incorporates a rubidium cell, rubidium lamp, and servo electronics to utilize the ground-state hyperfine transition of the rubidium atom, at approximately 6.834x GHz. The VCXO is locked to the rubidium atomic resonance in the following manner. The VCXO frequency of 50.255x MHz is an exact sub-multiple (÷136) of the atomic resonance frequency at 6.834x GHz. A microwave signal, having a frequency in the vicinity of 6.834x GHz, is generated from the nominal 50.255x MHz VCXO input. This microwave signal is used to resonate vaporized rubidium atoms within a sealed glass Rb resonance cell that is placed in a low
- Q microwave cavity. The microwave frequency generation method is designed so that the VCXO frequency is exactly 50.255x MHz when the microwave frequency is exactly equal to 6.834x GHz. The frequency of the signal applied to the microwave cavity can be maintained equal to 6.834x GHz by generating an error signal when the frequency varies, and using this error signal to servo the VCXO via its control voltage. The error signal is generated in the physics package. Light from the rubidium lamp, produced by an excited plasma discharge, is filtered and passed through the rubidium resonance cell where it interacts with rubidium atoms in the vapor. After passing through the resonance cell, this light is incident upon a photocell. When the applied microwave frequency is equal to 6.834x GHz, the rubidium atoms are resonated by the microwave field in the cavity; this causes the light reaching the photocell to decrease. The decrease in light, when the microwave frequency is equal to the sharply defined Rubidium frequency, is then converted electronically to an error signal with phase and amplitude information that is used to steer the VCXO via its control voltage and keep it on frequency at 50.255x



Rubidium Standard FE 5680A .. Tripler ... GALI-39 Wideband Amplifier .. Splitter.

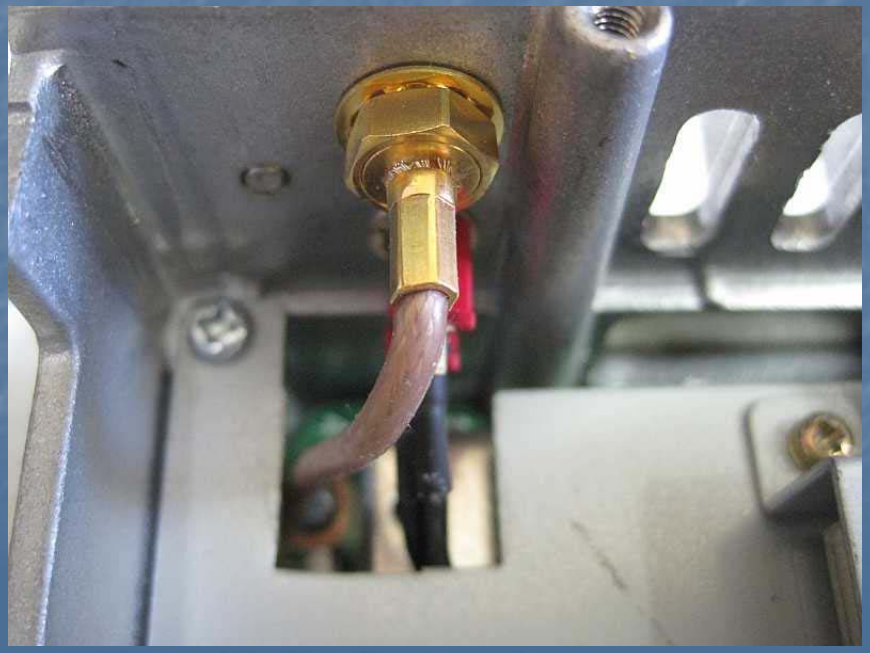
Power Supply

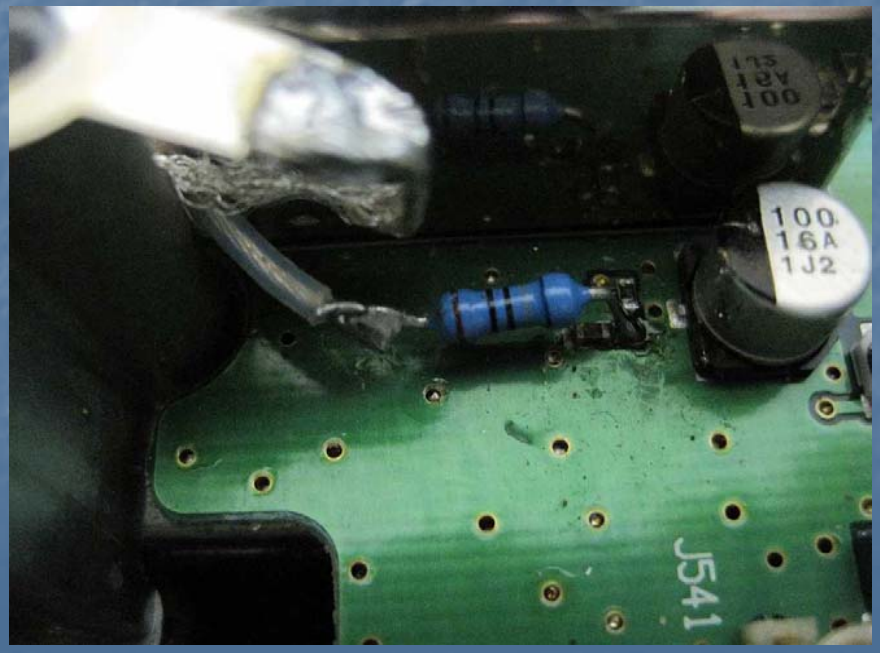
RUBIDIUM STANDARD
30.200.000 Hz
Lock
VK3UM
Suite 10, Victoria Square, 857 Doncaster Road, Doncaster East, VIC 3109


12:07:20 29 MAR 2011
REF 20.0 dBm
AT 30 dB
MKR 30.2005 MHz
10.12 dBm
PEAK
LOG
10
dB/
SPAN
100.0 kHz
MARKER
+ CF
MARKER
NEXT
PEAK
NEXT PK
RIGHT
NEXT PK
LEFT
More
1 of 2
CENTER 30.2000 MHz
RES BW 1.0 kHz
VBW 1 kHz
SPAN 100.0 kHz
SWP 300 msec

RUBIDIUM STANDARD
+5v
+12v
+15v
Power
On
POWER SUPPLY

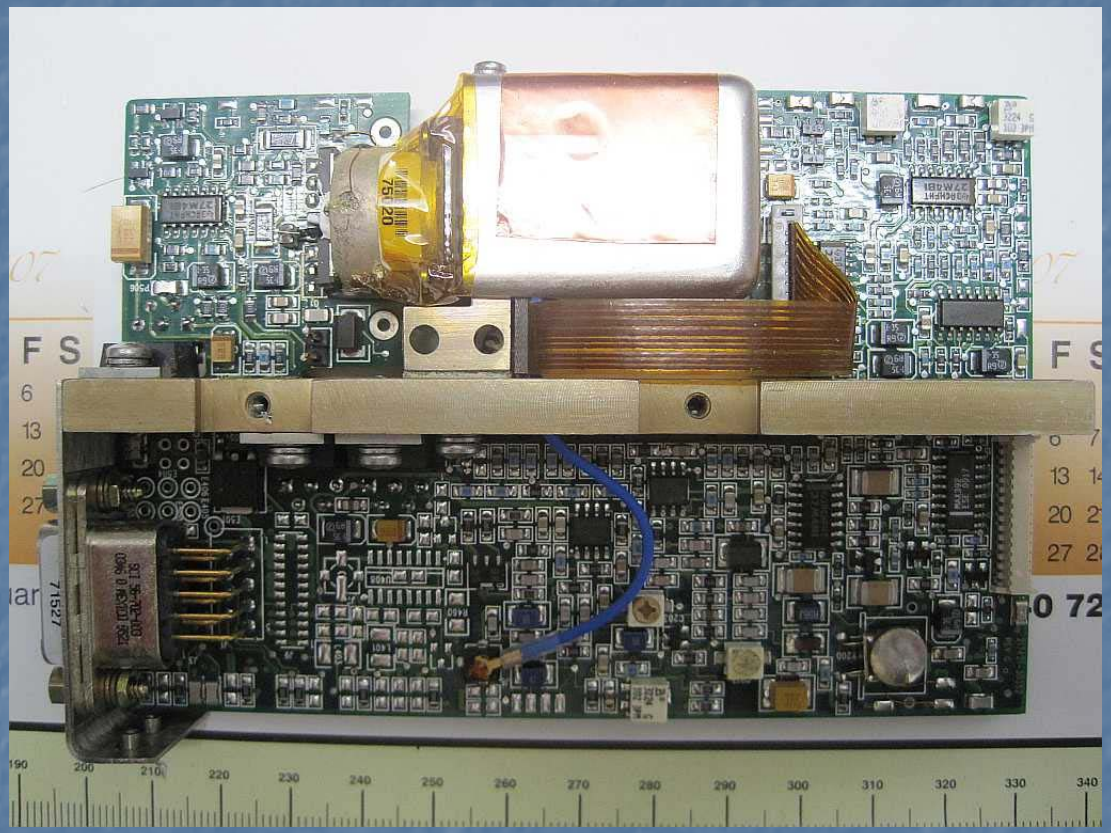
TRAKTON
RUBIDIUM STANDARD
Lock
VK3UM
RUBIDIUM STANDARD
POWER SUPPLY





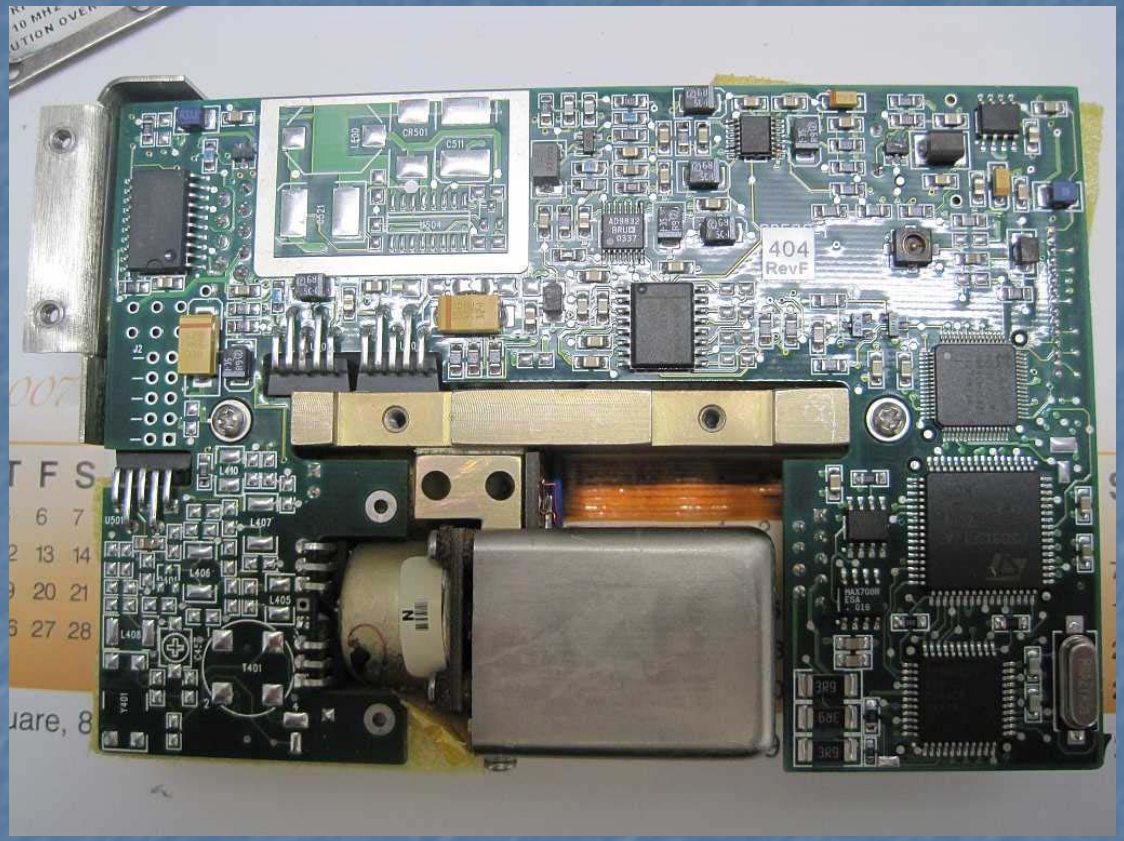
Modifications to the IC 910H

Inside the FE 5680A



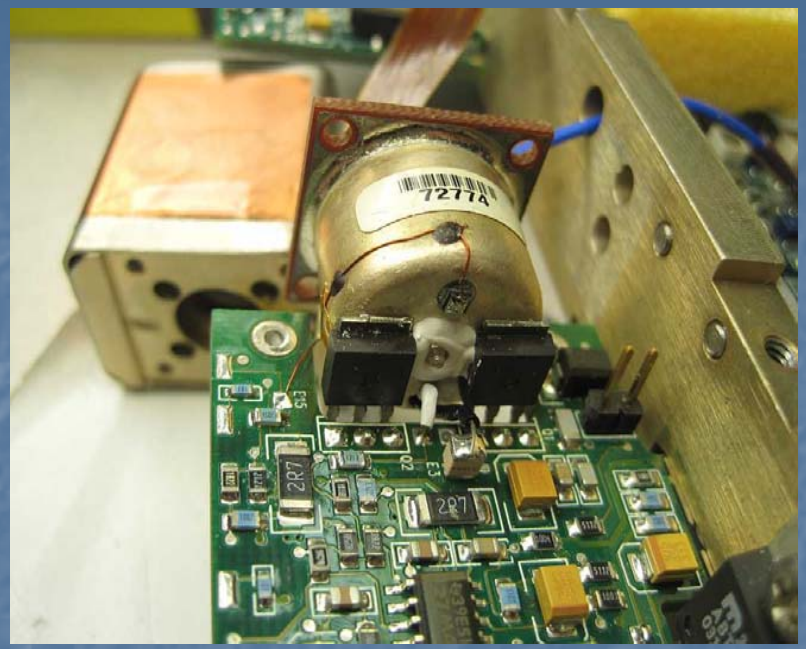
Top View

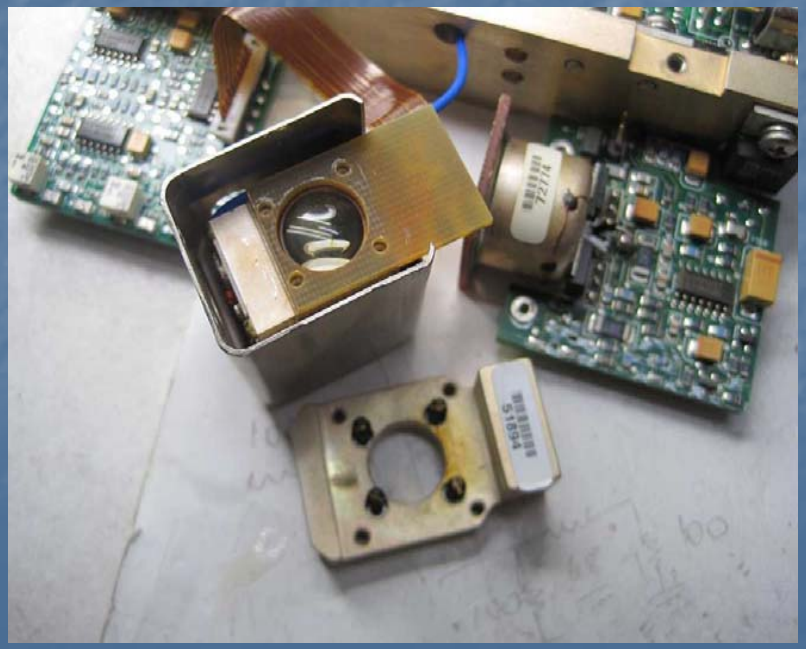
Inside the FE 5680A

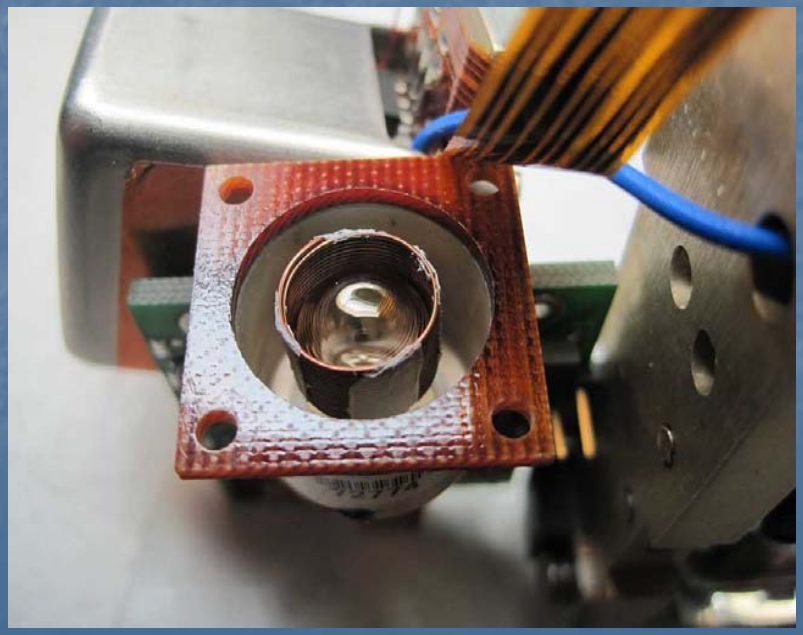


Bottom View

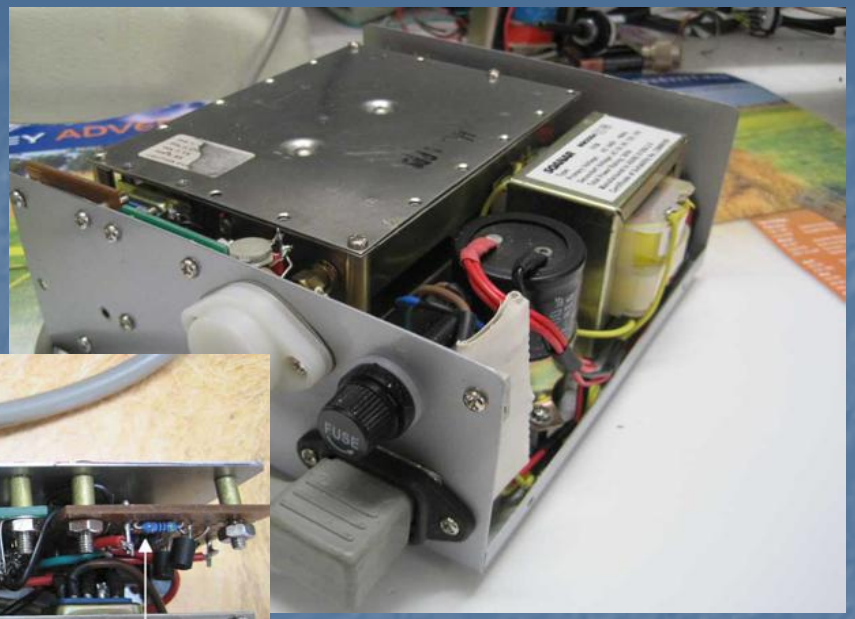
Rubidium Cell







Dual Standard Test Module



- Additional information can be obtained fro my web site at www.vk3um.com

- Acknowledgements
  - Tim Tuck VK2XTT
  - Murray Greenman ZL1BPU
  - Matthias Bopp DD1US
  - Bob Campbell VK4XV